

Bash Script Lab 1

System Administration Alexandria

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Using sed utility

- 1- Display the lines that contain the word “lp” in /etc/passwd file.
- 2- Display /etc/passwd file except the third line.
- 3- Display /etc/passwd file except the last line.
- 4- Display /etc/passwd file except the lines that contain the word “lp”.
- 5- Substitute all the words that contain “lp” with “mylp” in /etc/passwd file.

```
root@worker:~  
[root@worker ~]# #Answer or Q1  
[root@worker ~]# sed -n '/lp/p' /etc/passwd  
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin  
flatpak:x:992:989:User for flatpak system helper:/:/sbin/nologin  
[root@worker ~]# #Answer of Q2  
[root@worker ~]# sed '3 d' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
bin:x:1:1:bin:/bin:/sbin/nologin  
adm:x:3:4:adm:/var/adm:/sbin/nologin  
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
```

```
root@worker:~  
[root@worker ~]# #Answer of Q3  
[root@worker ~]# sed '$ d' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
bin:x:1:1:bin:/bin:/sbin/nologin  
daemon:x:2:2:daemon:/sbin:/sbin/nologin
```

```
root@worker:~  
[root@worker ~]# #Answer of Q4  
[root@worker ~]# sed '/lp/d' /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
bin:x:1:1:bin:/bin:/sbin/nologin  
daemon:x:2:2:daemon:/sbin:/sbin/nologin
```

```
root@worker:~  
[root@worker ~]# #Answer of Q5  
[root@worker ~]# sed -n 's/lp/mylp/p' /etc/passwd  
mylp:x:4:7:lp:/var/spool/lpd:/sbin/nologin  
flatpak:x:992:989:User for flatpak system hemylper:/:/sbin/nologin  
[root@worker ~]#
```

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System Administration Alexandria

Using awk utility

- 1- Print full name (comment) of all users in the system.
- 2- Print login, full name (comment) and home directory of all users.(Print each line preceded by a line number)
- 3- Print login, uid and full name (comment) of those uid is greater than 500
- 4- Print login, uid and full name (comment) of those uid is exactly 500
- 5- Print line from 5 to 15 from /etc/passwd
- 6- Change lp to mylp
- 7- Print all information about greatest uid.
- 8- Get the sum of all accounts id's.

```
root@worker:~  
[root@worker ~]# # Answer of Q1  
[root@worker ~]# awk 'BEGIN {FS=":"} {print $5}' /etc/passwd  
root  
bin  
daemon  
adm  
lp
```

```
root@worker:~  
[root@worker ~]# #Answer of Q2  
[root@worker ~]# awk 'BEGIN {FS=":";OFS=" | "} {print NR, $1, $5, $6}' /etc/passwd  
1 | root | root | /root  
2 | bin | bin | /bin  
3 | daemon | daemon | /sbin  
4 | adm | adm | /var/adm
```

```
root@worker:~  
[root@worker ~]# #Answer of Q3  
[root@worker ~]# awk 'BEGIN {FS=":";OFS=" | "} {if ($3 > 500) { print $1, $3, $5}}' /etc/passwd  
nobody | 65534 | Kernel Overflow User  
systemd-coredump | 999 | systemd Core Dumper  
polkitd | 998 | User for polkitd  
colord | 997 | User for colord  
pipewire | 996 | PipeWire System Daemon
```

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```
root@worker:~  
[root@worker ~]# #Answer of Question 4  
[root@worker ~]# id 500  
uid=500(testuser) gid=1015(testuser) groups=1015(testuser)  
[root@worker ~]# awk 'BEGIN {FS=":";OFS=" | "} {if ($3 == 500) { print $1, $3, $5}}'  
' /etc/passwd  
testuser | 500 |  
[root@worker ~]#  
  
root@worker:~  
[root@worker ~]# #Answer of Question 5  
[root@worker ~]# awk '{ if (NR >= 5 && NR <= 15) {print $0} }' /etc/passwd  
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin  
sync:x:5:0:sync:/sbin:/bin/sync  
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown  
halt:x:7:0:halt:/sbin:/sbin/halt  
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin  
operator:x:11:0:operator:/root:/sbin/nologin  
games:x:12:100:games:/usr/games:/sbin/nologin  
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin  
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin  
tss:x:59:59:Account used for TPM access:/:/sbin/nologin  
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin  
[root@worker ~]#  
  
root@worker:~  
[root@worker ~]# #Answer of Question 6  
[root@worker ~]# awk 'BEGIN{FS="lp"; OFS="mylb"} /lp/ { $1=$1; print $0 }' /etc/passwd  
mylb:x:4:7:mylb:/var/spool/mylbd:/sbin/nologin  
flatpak:x:992:989:User for flatpak system hemylber:/:/sbin/nologin  
[root@worker ~]#  
  
root@worker:~  
[root@worker ~]# #Answer of Question 7  
[root@worker ~]# awk 'BEGIN {FS=":";max=0;line=""} {if ($3>max) {max=$3 ; line=$0} } END{print line}' /etc/passwd  
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin  
[root@worker ~]#  
  
root@worker:~  
[root@worker ~]# #Answer of Question 8  
[root@worker ~]# awk 'BEGIN {FS=":"}{sum += $3} END {print sum}' /etc/passwd  
92734  
[root@worker ~]#
```